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MILITARY SPECIFICATION

PUMP UNIT, LIQUID-VACUUM, POWER DRIVEN

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification establishes requirements for one type of power driven liquid-vacuum pump unit.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal forms a part of this specification to the extent specified herein:

STANDARDS

FEDERAL

FED-STD-123 - Marking for Domestic Shipment (Civil Agencies).

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-252 - Wired Equipment, Classification of Visual and Mechanical Defects for.

FSC 6665

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-810 - Environmental Test Methods.
MS17821 - Strap, Cable, Adjustable, Self-Clinching, Plastic, Type 1, Class 1.

DRAWINGS AND TECHNICAL DATA PACKAGE LIST (TDPL)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

TDPL 5-15-4600 - Pump Liquid-Vacuum, Power Driven.
D136-44-176 - Tester, Functional, Liquid-Vacuum, Pump Unit, Power Driven, Q196.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on TDPL 5-15-4600 and on the subsidiary drawings shall conform to the specifications listed thereon and to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the pump unit shall conform to the specifications and drawings listed on TDPL 5-15-4600 and subsidiary drawings.

3.2 Manufacture and assembly. The pump unit shall be manufactured and assembled as specified on Drawing E5-15-4600.

3.3 Air system leakage. The static air leak rate of the pump unit air system shall not exceed 2.0 cubic centimeters per minute with the internal air pressure of the pump unit air system 49.0 ± 1.0 inch (1245 ± 25 millimeters (mm)) of water less than that outside of the pump unit when tested as specified in 4.4.4.1.

3.4 Liquid system leakage. The static air leakage rate of the pump unit liquid system shall not exceed 0.5 cubic centimeter per minute with the internal air pressure of the pump unit liquid system 49.0 ± 1.0 inch (1245 ± 25 millimeters) of water greater than that outside of the pump unit when tested as specified in 4.4.4.2.

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3.5 Simulated alarm. With the pump unit energized by 24.0 ± 1.0 volt direct current applied at pump unit connector pins A positive and C negative and the pump unit motor running at 2900 ± 200 revolutions per minute, connection of the minus terminal of the power source to connector pin B shall cause the pump unit motor to stop when tested as specified in 4.4.4.3.

3.6 Liquid pump pressure capability. When tested as specified in 4.4.4.4, the pressure capability of the liquid pump shall be no less than 6.0 nor more than 30.0 pounds per square inch gauge (41.4 nor more than 206.8 kilopascals).

3.7 Performance. The performance characteristics of the pump unit shall be as specified in table I when tested as described in 4.4.4.5 under the test conditions listed below:

(a) Applied voltages of 18.5 (+0.5 or -0.0) and 36.0 ± 0.5 volt direct current, except that liquid flow need not be measured at 36.0 volts.

(b) Air pump inlet head of 6.0 ± 1.0 inch (152 ± 25 mm) of water.

(c) Air pump outlet head of 0.5 ± 0.1 inch (12.7 ± 2.5 mm) of water.

Table I. Performance characteristics

Characteristics	Requirement
Motor speed	2,900 to 3,100 revolutions per minute
Current	60.0 milliamperes, maximum
Airflow at air pump inlet	1.1 to 1.5 liters per minute
Liquid flow through liquid pump at 18.5 (+0.5 or -0.0) volt only.	0.15 to 0.23 milliliter per minute

3.8 Operation at low temperature after cold storage. The pump unit shall meet the requirements of 3.7 after being stored at -40° ± 5°F (-40.0° ± 2.8°C) for 4 hours and then allowed to stabilize at a temperature of +40° ± 5°F (4.4° ± 2.8°C) for a minimum of 3 hours when tested as specified in 4.3.3.1.

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3.9 Operation at high temperature. The pump unit shall meet the requirements of 3.7 after being stored at a temperature of 120° ± 5°F (48.9° ± 2.8°C) for a minimum of 3 hours when tested as specified in 4.3.3.2.

3.10 Operation after vibration. The pump unit shall meet the requirements of 3.7 after being subjected to the vibration environment described in MIL-STD-810, method 514, procedure VIII, curve Y (6,000 miles) (9,656 kilometers) when tested as specified in 4.3.3.3. There shall be no evidence of physical damage or loose parts.

3.11 Preproduction. Prior to the start of regular production, a preproduction sample of pump units shall be produced in accordance with this specification for examination and test (see 4.3).

3.12 Workmanship. The workmanship of the pump unit shall be in accordance with MIL-STD-252.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

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4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of eight pump units shall be produced using the same materials, methods, equipment, and processes as will be used during regular production.

4.3.2 Inspection procedures.

4.3.2.1 Examination. The preproduction sample of pump units shall be examined in accordance with the classification of defects (4.4.3.3).

4.3.2.2 Test. The preproduction sample of pump units shall be tested in accordance with 4.3.3 and 4.4.4.

4.3.3 Tests. Tests shall be conducted as follows:

4.3.3.1 Operation at low temperature after cold storage. Place the pump unit into a temperature test chamber already at -40° ± 5°F (-40° ± 2.8°C) and permit it to remain undisturbed at that temperature for 4 hours. Increase the temperature in the chamber to the higher temperature re quired by 3.8 and allow to stabilize at that temperature for a minimum of 3 hours. Then remove the pump unit from the test chamber and quickly subject it to the performance test specified in 4.4.4.5. The motor speed, current, and airflow portions of the performance test must be completed within 8 minutes after the pump unit was removed from the temperature test chamber. All of the pump unit performance characteristics must be as required by 3.7.

4.3.3.2 Operation at high temperature. Place the pump unit into a temperature test chamber already at the temperature required by 3.9, and permit it to remain undisturbed at that temperature for at least 3 hours. Then, remove the pump unit from the test chamber and quickly subject it to the performance test of 4.4.4.5. The motor speed, current, and airflow portions of the performance test must be completed within 8 minutes after the pump unit was removed from the temperature test chamber. All of the pump unit performance characteristics must be as required by 3.7.

4.3.3.3 Operation after vibration. Mount the pump unit onto a vibration exciter. No electrical, pneumatic, or liquid connections need be made; the pump unit need not be operated while it is on the vibration exciter. Subject the pump unit to the vibration environment required by 3.10. Then, remove the pump unit from the vibration exciter and subject it to the performance test of 4.4.4.5. All of the pump unit performance characteristics must be as required by 3.7, and check for physical damage, loose nuts, bolts or parts.

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4.3.4 Acceptance/rejection criteria. The preproduction sample shall comply with the examinations and tests specified in 4.3.2 to be acceptable. The contractor shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the pump units offered for acceptance at one time, which have been produced by one manufacturer, at one plant, under essentially the same manufacturing conditions.

4.4.2 Sampling for examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.3 Inspection procedures.

4.4.3.1 For examination. Sample pump units shall be examined in accordance with the classification of defects and MIL-STD-105.

4.4.3.2 For test. Each pump unit in the lot shall be tested in accordance with 4.4.4.1 through 4.4.4.5. After completion of the tests, the pump unit shall be flushed as specified in 4.4.4.6.

4.4.3.3 Classification of defects.

(a) Pump unit, liquid-vacuum, power driven (Drawing E5-15-4600).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 1.0 percent defective	
101	Workmanship (applicable defects listed as Major in MIL-STD-252, see 3.12)	
102	Spatial relationship of pump air and liquid connections to the electrical connector	Dwg C5-15-5381 Jig Assembly
103	Height of tube relaxing pin above pump chassis	

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4.4.3.3 Classification of defects (continued).

Minor:	AQL 4.0 percent defective	
201	Workmanship (applicable defects listed as Minor in MIL-STD-252, see 3.12)	
	AQL 6.5 percent defective	
202	Workmanship (applicable defects listed as Control in MIL-STD-252, see 3.12)	

4.4.4 Tests. Tests shall be performed and the pump unit flushed as follows:

4.4.4.1 Air system leakage. Determine the air system leakage utilizing the Q196 (D136-44-176) in accordance with the instruction manual.

4.4.4.2 Liquid system leakage. Determine the liquid system leakage utilizing the Q196 in accordance with the instruction manual.

4.4.4.3 Simulated alarm. Test response to a simulated alarm by utilizing the Q196 in accordance with the instruction manual.

4.4.4.4 Liquid pump pressure capability. Determine the liquid pump pressure capability utilizing the Q196 in accordance with the instruction manual.

4.4.4.5 Performance. Determine the performance characteristics of the pump unit utilizing the Q196 in accordance with the instruction manual.

4.4.4.6 Flushing procedure. Flush the liquid pump of each pump unit with deionized water at a maximum pressure of 45.0 inches of water above atmospheric pressure. After flushing, blow-out the liquid pump with filtered ambient air at a maximum pressure of 45.0 inches (1143 mm) of water above atmospheric pressure. Examine the air inlet and air outlet ports of the air pump for the presence of oxime solution residues. If such residues are found, clean the air pump by flushing with deionized water. Force the flushing water into the air inlet port of the air pump at a maximum pressure of 45.0 inches of water above atmospheric pressure. After flushing, blow-out the air pump with filtered ambient air at a maximum pressure of 45.0 inches of water above atmospheric pressure. This air must enter the air pump at the inlet port.

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5. PACKAGING

5.1 Unit packing, packing and marking, interplant shipment (see 6.3). The tube guide shall be secured as specified on figure 1. The pump unit shall be unit packed and packed to provide adequate protection from physical damage from the supply source to the first receiving activity for immediate use or further processing. Shipping containers shall be in compliance with the rules and regulations applicable to the mode of transportation. Marking shall be in conformance with Fed-Std-123 and shall include the contractors name, address and contract number.

5.2 Repair parts data sheet. When this item is being procured for storage and issue as a repair part, preservation, unit packing, packing and marking shall be as specified on the repair parts packaging data sheet which is identified by its National stock number.

6. NOTES

6.1 Intended use. This pump unit assembly is used to complete a function as a subassembly in the detector unit which is used in a series of chemical agent alarms.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preproduction.

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

6.3 Interplant shipment. Packaging and packing for interplant shipment is for supplies and material that do not enter the military supply system. Typical interplant shipments are shipments from a vendor to a contractor, or from a contractor to a military arsenal or plant.

6.4 International interest. Certain provisions of this specification are the subject of international standardization agreement QSTAG 251. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned,

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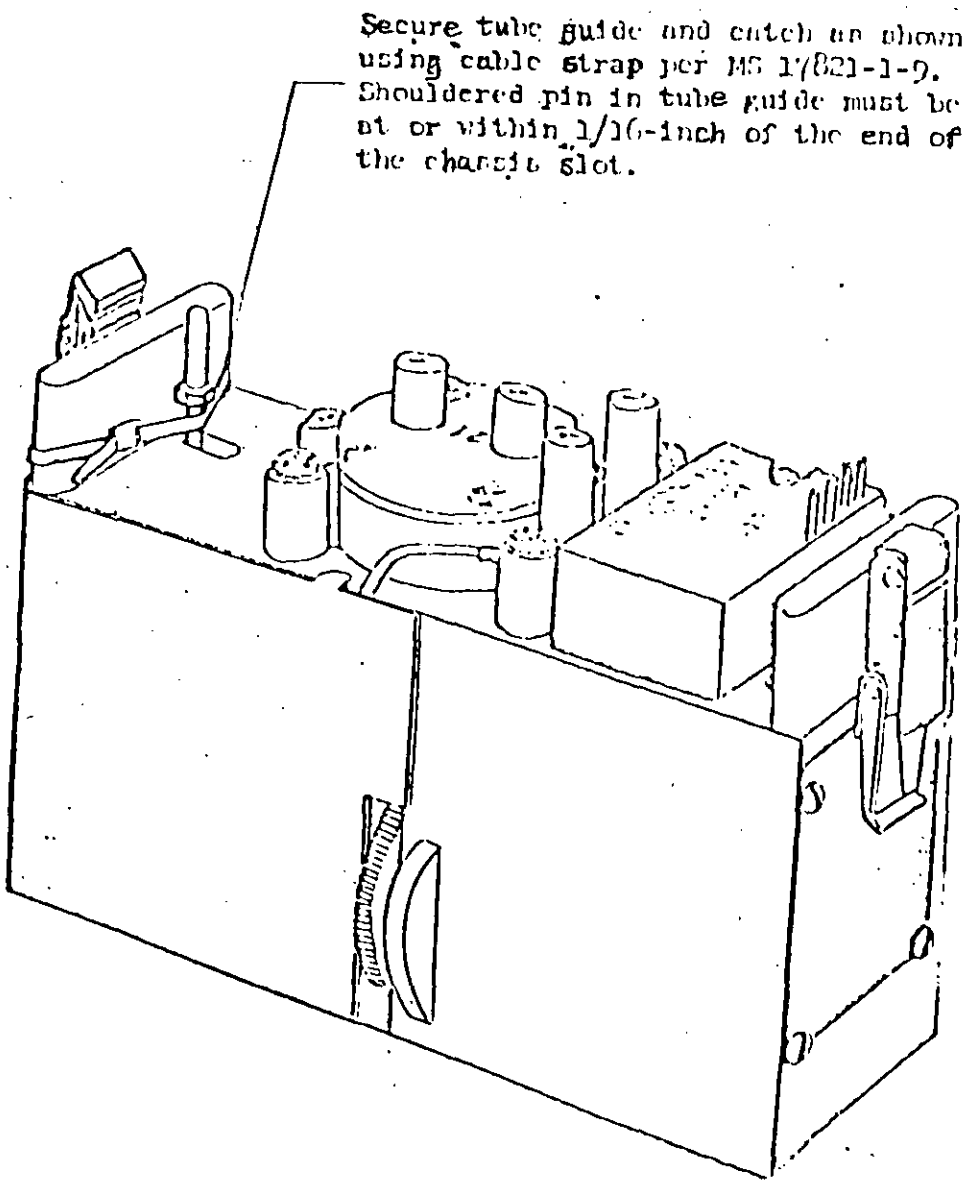


Figure 1. Securing of tube guide

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the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

Custodian:

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Preparing activity:

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